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NEET · PHYSICS · CHAPTER-WISE PYQS

# Magnetism and Matter

NEET Physics · Previous Year Questions with Answer Key

| 47 | 29 | 2001 – 2024 | 1 | English |
|---|---|---|---|---|
| QUESTIONS | PAPERS COVERED | YEARS | OUT OF SYLLABUS | LANGUAGE |

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## MCQ (Single Correct Answer) 46 questions

**Q1.** NEET 2024 (Re-Examination)

The magnetic potential energy, when a magnetic bar of magnetic moment $\vec{m}$ is placed perpendicular to the magnetic field $\vec{B}$ is

(A) $-\frac{mB}{2}$ (B) Zero (C) $-mB$ (D) $mB$

Open solution

**Q2.** NEET 2024 (Re-Examination)

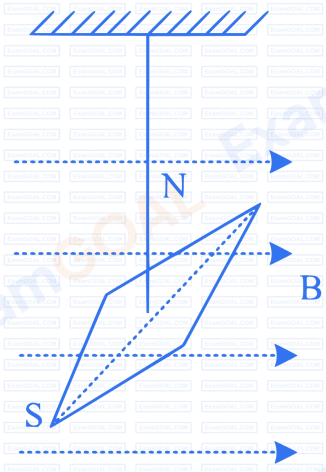
The incorrect relation for a diamagnetic material (all the symbols carry their usual meaning and $\varepsilon$ is a small positive number) is

(A) $\mu < \mu_0$ (B) $0 \leq \mu_r < 1$ (C) $-1 \leq \chi < 0$ (D) $1 < \mu_r < 1 + \varepsilon$

Open solution

**Q3.** NEET 2024 (Re-Examination)

The magnetic moment and moment of inertia of a magnetic needle as shown are, respectively, $1.0 \times 10^{-2}\ \mathrm{A\,m^2}$ and $\frac{10^{-6}}{\pi^2}\ \mathrm{kg\,m^2}$. If it completes 10 oscillations in $10\ \mathrm{s}$, the magnitude of the magnetic field is




(A) 0.4 T (B) 4 T (C) 0.4 mT (D) 4 mT

Open solution

**Q4.** NEET 2024 (Re-Examination)

The magnetic moment of an iron bar is $M$. It is now bent in such a way that it forms an arc section of a circle subtending an angle of $60^\circ$ at the centre. The magnetic moment of this arc section is

(A) $\frac{3M}{\pi}$ (B) $\frac{4M}{\pi}$ (C) $\frac{M}{\pi}$ (D) $\frac{2M}{\pi}$

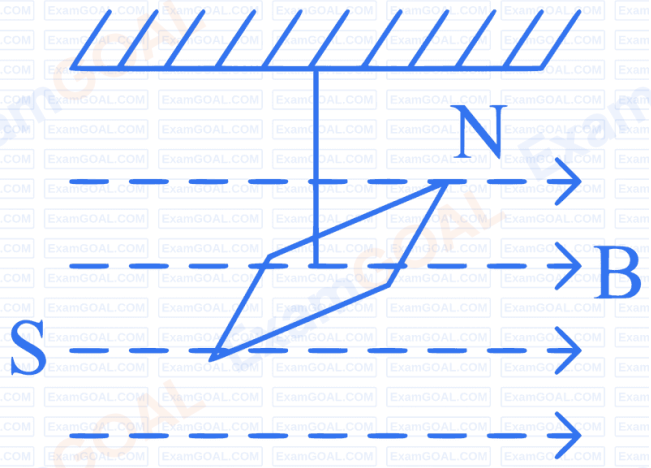
Open solution

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**Q5.** NEET 2024

Open solution

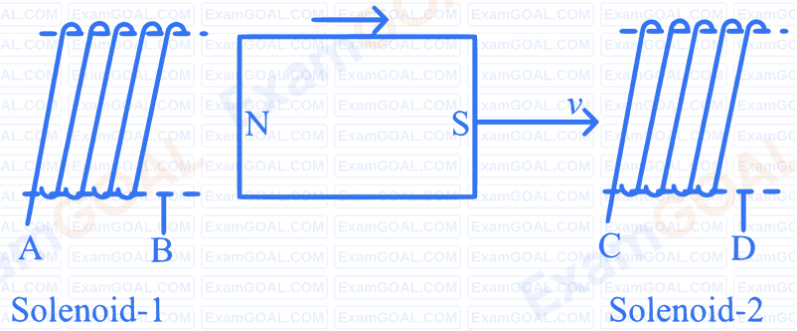
In a uniform magnetic field of $0.049\ \mathrm{T}$, a magnetic needle performs 20 complete oscillations in 5 seconds as shown. The moment of inertia of the needle is $9.8 \times 10^{-6}\ \mathrm{kg\,m^2}$. If the magnitude of magnetic moment of the needle is $x \times 10^{-5}\ \mathrm{Am^2}$, then the value of '$x$' is :




(A) $5\pi^2$ (B) $128\pi^2$ (C) $50\pi^2$ (D) $1280\pi^2$

**Q6.** NEET 2024

Open solution




In the above diagram, a strong bar magnet is moving towards solenoid-2 from solenoid-1. The direction of induced current in solenoid-1 and that in solenoid-2, respectively, are through the directions:

(A) AB and DC (B) BA and CD (C) AB and CD (D) BA and DC

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**Q7.** NEET 2024

Open solution

Match List I with List II

| | List I (Material) | | List II (Susceptibility ($$\chi)) |
|---|---|---|---|
| A. | Diamagnetic | I. | $\chi = 0$ |
| B. | Ferromagnetic | II. | $0 > \chi \geq -1$ |
| C. | Paramagnetic | III. | $\chi >> 1$ |
| D. | Non-magnetic | IV. | $0 < \chi < \varepsilon$ (a small positive number) |

Choose the correct answer from the options given below.

**(A)** A-II, B-III, C-IV, D-I

**(B)** A-II, B-I, C-III, D-IV

**(C)** A-III, B-II, C-I, D-IV

**(D)** A-IV, B-III, C-II, D-I

**Q8.** NEET 2024

Open solution

A sheet is placed on a horizontal surface in front of a strong magnetic pole. A force is needed to:

A. hold the sheet there if it is magnetic.

B. hold the sheet there if it is non-magnetic.

C. move the sheet away from the pole with uniform velocity if it is conducting.

D. move the sheet away from the pole with uniform velocity if it is both, non-conducting and non-polar.

Choose the correct statement(s) from the options given below:

**(A)** B and D only **(B)** A and C only **(C)** A, C and D only **(D)** C only

**Q9.** NEET 2024

Open solution

An iron bar of length $L$ has magnetic moment $M$. It is bent at the middle of its length such that the two arms make an angle $60^\circ$ with each other. The magnetic moment of this new magnet is :

**(A)** $M$ **(B)** $\frac{M}{2}$ **(C)** $2M$ **(D)** $\frac{M}{\sqrt{3}}$

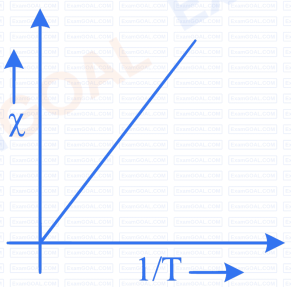
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**Q10.** NEET 2023 Manipur

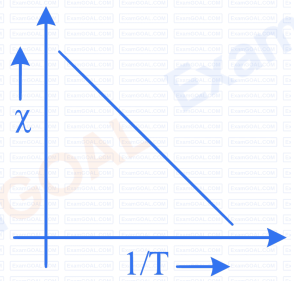
The variation of susceptibility $(\chi)$ with absolute temperature ( $\mathrm{T}$ ) for a paramagnetic material is represented as :

Open solution

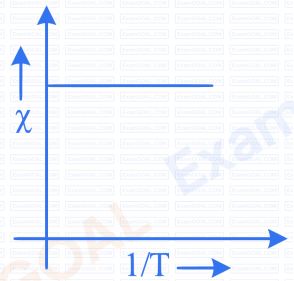
(A)



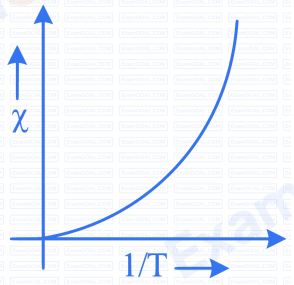

(B)




(C)




(D)




**Q11.** NEET 2022 Phase 1

A big circular coil of 1000 turns and average radius 10 m is rotating about its horizontal diameter at 2 rad $s^{-1}$. If the vertical component of earth's magnetic field at that place is $2 \times 10^{-5}$ T and electrical resistance of the coil is 12.56 $\Omega$, then the maximum induced current in the coil will be

Open solution

(A) 0.25 A (B) 1.5 A (C) 1 A (D) 2 A

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**Q12.** NEET 2021

Open solution

In the product

$\overrightarrow{F} = q\left(\overrightarrow{v} \times \overrightarrow{B}\right)$

$= q\overrightarrow{v} \times \left(B\hat{i} + B\hat{j} + B_0\hat{k}\right)$

For q = 1 and $\overrightarrow{v} = 2\hat{i} + 4\hat{j} + 6\hat{k}$ and $\overrightarrow{F} = 4\hat{i} - 20\hat{j} + 12\hat{k}$

What will be the complete expression for $\overrightarrow{B}$ ?

(A) $6\hat{i} + 6\hat{j} - 8\hat{k}$

(B) $-8\hat{i} - 8\hat{j} - 6\hat{k}$

(C) $-6\hat{i} - 6\hat{j} - 8\hat{k}$

(D) $8\hat{i} + 8\hat{j} - 6\hat{k}$

**Q13.** NEET 2021

Open solution

A uniform conducting wire of length 12a and resistance 'R' is wound up as a current carrying coil in the shape of ,

(i) an equilateral triangle of side 'a'

(ii) A square of side 'a'

The magnetic dipole moments of the coil in each case respectively are :

(A) $4Ia^2$ and $3Ia^2$

(B) $\sqrt{3}\ Ia^2$ and $3I\ a^2$

(C) $3Ia^2$ and $Ia^2$

(D) $3Ia^2$ and $4Ia^2$

**Q14.** NEET 2020 Phase 1

Open solution

An iron rod of susceptibility 599 is subjected to a magnetizing field of 1200 A $m^{-1}$. The permeability of the material of the rod is :
$(\mu_0 = 4\pi \times 10^{-7} TmA^{-1})$

(A) $8.0 \times 10^{-5} TmA^{-1}$

(B) $2.4\pi \times 10^{-5} TmA^{-1}$

(C) $2.4\pi \times 10^{-7} TmA^{-1}$

(D) $2.4\pi \times 10^{-4} TmA^{-1}$

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**Q15.** NEET 2018

A thin diamagnetic rod is placed vertically between the poles of an electromagnet. When the current in the electromagnet is switched on, then the diamagnetic rod is pushed up, out of the horizontal magnetic field. Hence the rod gains gravitational potential energy. The work required to do this comes from

(A) The current source

(B) The magnetic field

(C) The lattice structure of the material of the rod

(D) The induced electric field due to the changing magnetic field

Open solution

**Q16.** NEET 2017

If $\theta_1$ and $\theta_2$ be the apparent angles of dip observed in two vertical planes at right angles to each other, then the true angle of dip $\theta$ is given by

(A) $\tan^2\theta = \tan^2\theta_1 + \tan^2\theta_2$

(B) $\cot^2\theta = \cot^2\theta_1 - \cot^2\theta_2$

(C) $\tan^2\theta = \tan^2\theta_1 - \tan^2\theta_2$

(D) $\cot^2\theta = \cot^2\theta_1 + \cot^2\theta_2$

Open solution

**Q17.** NEET 2017

A 250-turn rectangular coil of length 2.1 cm and width 1.25 cm carries a current of 85 $\mu$A and subjected to a magnetic field of strength 0.85 T. Work done for rotating the coil by 180$^o$ against the torque is

(A) 4.55 $\mu$J (B) 2.3 $\mu$J (C) 1.15 $\mu$J (D) 9.1 $\mu$J

Open solution

**Q18.** NEET 2016 Phase 2

A bar magnet is hung by a thin cotton thread in a uniform horizontal magnetic field and is in equilibrium state. The energy required to rotate it by 60$^o$ is W. Now the torque required to keep the magnet in this new position is

(A) $\frac{W}{\sqrt{3}}$ (B) $\sqrt{3}W$ (C) $\frac{\sqrt{3}W}{2}$ (D) $\frac{2W}{\sqrt{3}}$

Open solution

**Q19.** NEET 2016 Phase 1

The magnetic susceptibility is negative for

(A) ferromagnetic material only

(B) paramagnetic and ferromagnetic materials

(C) diamagnetic material only

(D) paramagnetic material only

Open solution

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**Q20.** AIPMT 2015

A rectangular coil of length 0.12 m and width 0.1 m having 50 turns of wire is suspended vertically in a uniform magnetic field of strength 0.2 weber/$m^2$. The coil carries a current of 2 A. If the plane of the coil is inclined at an angle of $30^o$ with the direction of the field, the torque required to keep the coil in stable equilibrium will be

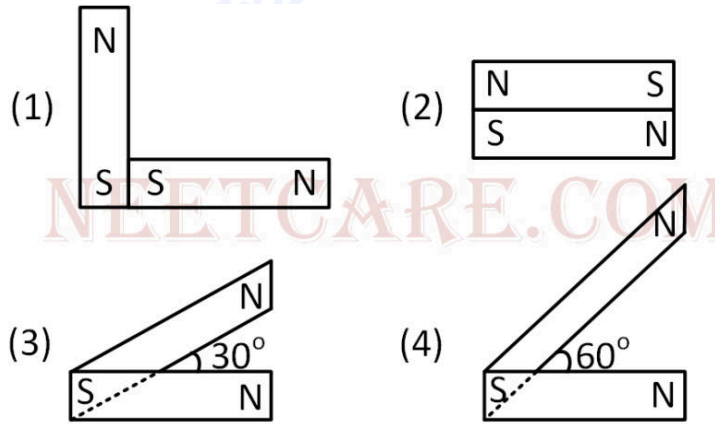
Open solution

(A) 0.24 Nm (B) 0.12 Nm (C) 0.15 Nm (D) 0.20 Nm

**Q21.** AIPMT 2014

Following figures show the arrangement of bar magnets in different configurations. Each magnet has magnetic dipole moment $\vec{m}$. Which configuration has highest net magnetic dipole moment ?

Open solution




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(A) (1) (B) (2) (C) (3) (D) (4)

**Q22.** NEET 2013 (Karnataka)

A bar magnet of magnetic moment M is placed at right angles to a magnetic induction B. If a force F is experienced by each pole of the magnet, the length of the magnet will be

Open solution

(A) MB/F (B) BF/M (C) MF/B (D) F/MB

**Q23.** NEET 2013

A current loop in a magnetic field

Open solution

(A) can be in equilibrium in two orientations, both the equilibrium states are unstable.

(B) can be in equilibrium in two orientations, one stable while the other is unstable.

(C) experiences a torque whether the field is uniform or non uniform in all orientations.

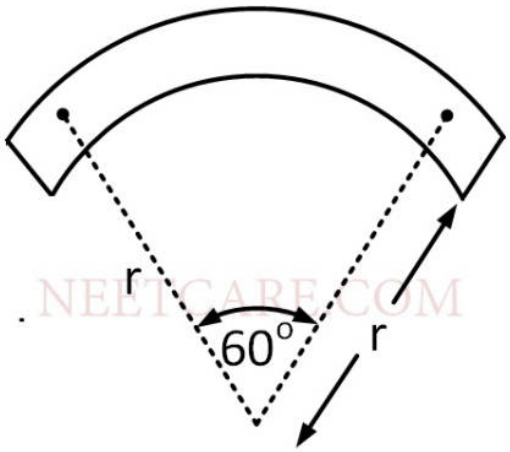
(D) can be in equilibrium in one orientation.

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**Q24.** NEET 2013

A bar magnet of length $'l'$ and magnetic dipole moment 'M' is bent in the form of an arc as shown in figure. The new magnetic dipole moment will be

Open solution




(A) $\frac{2}{\pi}M$ (B) $\frac{M}{2}$ (C) M (D) $\frac{3}{\pi}M$

**Q25.** AIPMT 2012 Mains

A magnetic needle suspended parallel to a magnetic field requires $\sqrt{3}J$ of work to turn it through $60^o$. The torque needed to maintain the needle in this position will be

(A) $2\sqrt{3}J$ (B) $3\ J$ (C) $\sqrt{3}\,J$ (D) $\frac{3}{2}\,J$

Open solution

**Q26.** AIPMT 2012 Prelims

A compass needle which is allowed to move in a horizontal plane is taken to a geomagnetic pole. It

(A) will become rigif showing no movement (B) will stay in any position

(C) will stay in north-south direction only (D) will stay in east-west direction only

Open solution

**Q27.** AIPMT 2011 Mains

A short bar magnet of magnetic moment 0.4 J $T^{-1}$ is placed in a uniform magnetic field of 0.16 T. The magnet is in stable equilibrium when the potential energy is

(A) 0.064 J (B) $-$ 0.064 J (C) zero (D) $-$ 0.082 J

Open solution

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**Q28.** AIPMT 2011 Prelims

There are four light-weight-rod samples A, B, C, D separately suspended by threads. A bar magnet is slowly brought near each sample and the following observations are noted

(i) A is feebly repelled

(ii) B is feebly attracted

(iii) C is strongly attracted

(iv) D remains unaffected

Which one of the following is true ?

(A) B is of a paramagnetic material

(B) C is of a diamagnetic material

(C) D is of a ferromagnetic material

(D) A is of a non-magnetic material

Open solution

**Q29.** AIPMT 2010 Mains

The magnetic moment of a diamagnetic atom is

(A) much greater than one

(B) 1

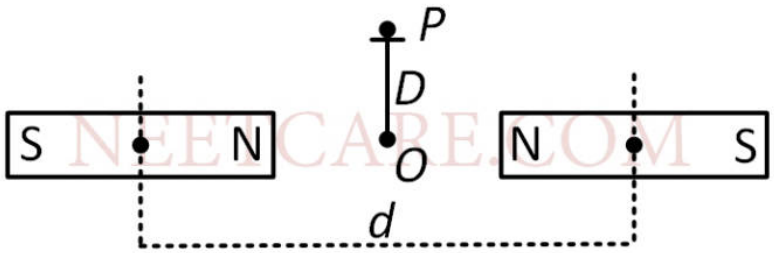
(C) between zero and one

(D) equal to zero

Open solution

**Q30.** AIPMT 2010 Mains

Two identical bar magnets are fixed with their centers at a distance d apart. A stationary charge Q is placed at P in between the gap of the two magnets at a distance D from the centre O as shown in the figure




The force on the charge Q is

(A) zero

(B) directed along OP

(C) directed along PO

(D) directed perpendicular to the plane of paper

Open solution

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**Q31.** AIPMT 2010 Prelims

Electromagnets are made of soft iron because soft iron has

(A) low retentivity and high coercive force
(B) high retentivity and high coercive force
(C) low retentivity and low coercive force
(D) high retentivity and low coercive force

Open solution

**Q32.** AIPMT 2010 Prelims

A vibration magnetometer placed in magnetic meridian has a small bar magnet. The magnet executes oscillations with a time period of 2 sec in earth's horizontal magnetic field of 24 microtesla. When a horizontal field of 18 microtesla is produced opposite to the earth's field by placing a current carrying wire, the new time period of magnet will be

(A) 1 s
(B) 2 s
(C) 3 s
(D) 4 s

Open solution

**Q33.** AIPMT 2009

If a diamagnetic substance is brought near the north or the south pole of a bar magnet, it is

(A) repelled by the north pole and attracted by the south pole
(B) attracted by the north pole and repelled by the south pole
(C) attracted by both the poles
(D) repelled by both the poles

Open solution

**Q34.** AIPMT 2009

A bar magnet having a magnetic moment of $2 \times 10^4$ J $T^{-1}$ is free to rotate in a horizontal plane. A horizontal magnetic field B = $6 \times 10^{-4}$ T exists in the space. The work done in taking the magnet slowly from a direction parallel to the field to a direction $60^o$ from the field is

(A) 12 J
(B) 6 J
(C) 2 J
(D) 0.6 J

Open solution

**Q35.** AIPMT 2008

Curie temperature above which

(A) paramagnetic material becomes ferromagnetic material
(B) ferromagnetic material becomes diamagnetic material
(C) ferromagnetic material becomes paramagnetic material
(D) paramagnetic material becomes diamagnetic material

Open solution

**Q36.** AIPMT 2007

Nickel shows ferromagnetic property at room temperature. If the temperature is increased beyond Curie temperature, then it will show

(A) anti ferromagnetism
(B) no magnetic property
(C) diamagnetism
(D) paramagnetism.

Open solution

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**Q37.** AIPMT 2007

A charged particle (charge q) is moving in a circle of radius R with uniform speed v. The associated magnetic moment $\mu$ is given by

(A) $qvR^2$ (B) $qvR^2/2$ (C) qvR (D) qvR/2

Open solution

**Q38.** AIPMT 2006

Curie temperature above which

(A) paramagnetic material becomes ferromagnetic material

(B) ferromagnetic material becomes diamagnetic material

(C) ferromagnetic material becomes paramagnetic material

(D) paramagnetic material becomes diamagnetic material

Open solution

**Q39.** AIPMT 2005

If the magnetic dipole moment of an atom of diamagnetic material, paramagnetic material and ferromagnetic material are denoted by $\mu_d$, $\mu_p$ and $\mu_f$ respectively, then

(A) $\mu_d = 0$ and $\mu_p \neq 0$ (B) $\mu_d \neq 0$ and $\mu_p = 0$

(C) $\mu_p = 0$ and $\mu_f \neq 0$ (D) $\mu_d \neq 0$ and $\mu_f \neq 0$.

Open solution

**Q40.** AIPMT 2005

A coil in the shape of an equilateral triangle of side $l$ is suspended between the pole pieces of a permanent magnet such that $\vec{B}$ is in plane of the coil. If due to a current i in the triangle a torque $\tau$ acts on it, the side $l$ of the triangle is

(A) $\frac{2}{\sqrt{3}}\left(\frac{\tau}{Bi}\right)$ (B) $2\left(\frac{\tau}{\sqrt{3}Bi}\right)^{1/2}$

(C) $\frac{2}{\sqrt{3}}\left(\frac{\tau}{Bi}\right)^{1/2}$ (D) $\frac{1}{\sqrt{3}}\frac{\tau}{Bi}$

Open solution

**Q41.** AIPMT 2003

A diamagnetic material in a magnetic field moves

(A) from stronger to the weaker parts of the field

(B) from weaker to the stronger parts of the field

(C) perpendicular to the field

(D) in none of the above directions

Open solution

**Q42.** AIPMT 2003

According to Curie's law, the magnetic susceptibility of a substance at an absolute temperature $T$ is proportional to

(A) $1/T$ (B) $T$ (C) $1/T^2$ (D) $T^2$

Open solution

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**Q43.** AIPMT 2003

A bar magnet is oscillating in the Earth's magnetic field with a period T. What happens to its period and motion if its mass is quadrupled?

(A) motion remains simple harmonic with time period = $T/2$

(B) motion remains S.H.M with time period = 2T

(C) motion remains S.H.M with time period = 4T

(D) motion remains S.H.M and period remains nearly constant

Open solution

**Q44.** AIPMT 2002

Two bar magnets having same geometry with magnetic moments M and 2M, are firstly placed in such a way that their similar poles are same side then its time period of oscillation is $T_1$. Now the polarity of one of the magnet is reversed then time period of oscillation is $T_2$, then

(A) $T_1 < T_2$ (B) $T_1 = T_2$ (C) $T_1 > T_2$ (D) $T_2 = \infty$

Open solution

**Q45.** AIPMT 2001

Among which the magnetic susceptibility does not depend on the temperature ?

(A) diamagnetism (B) paramagnetism (C) ferromagnetism (D) ferrite.

Open solution

**Q46.** AIPMT 2001

Tangent galvanometer is used to measure

(A) potential difference (B) current

(C) resistance (D) charge

Open solution

## Out of syllabus 1 questions

*These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.*

**Q47.** NEET 2019 · MCQ (Single Correct Answer)

At a point A on the earth's surface the angle of dip, $\delta = +25^{o}$. At a point B on the earth's surface the angle of dip, $\delta = -25^{o}$. We can interpret that :

(A) A is located in the southern hemisphere and B is located in the northern hemisphere.

(B) A is located in the northern hemisphere and B is located in the southern hemisphere.

(C) A and B are both located in the southern hemisphere.

(D) A and B are both located in the northern hemisphere.

Open solution

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# Answer Key

## MCQ (Single Correct Answer)

| | | | | |
|---|---|---|---|---|
| 1. **B** | 2. **D** | 3. **C** | 4. **A** | 5. **D** |
| 6. **A** | 7. **A** | 8. **B** | 9. **B** | 10. **A** |
| 11. **C** | 12. **C** | 13. **B** | 14. **D** | 15. **A** |
| 16. **D** | 17. **D** | 18. **B** | 19. **C** | 20. **D** |
| 21. **C** | 22. **A** | 23. **B** | 24. **D** | 25. **B** |
| 26. **B** | 27. **B** | 28. **A** | 29. **D** | 30. **A** |
| 31. **C** | 32. **D** | 33. **D** | 34. **B** | 35. **C** |
| 36. **D** | 37. **D** | 38. **C** | 39. **A** | 40. **B** |
| 41. **A** | 42. **A** | 43. **B** | 44. **A** | 45. **A** |
| 46. **B** | | | | |

## Out of syllabus

| |
|---|
| 47. **B** |

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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